

# VAIBHAV VIKAS RANJAN

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## Education

International Institute of Information Technology, IIIT-NR

B.Tech Data Science and Artificial Intelligence

Nov 2022 - July 2026

Naya Raipur, Chhattisgarh

## Experience

Accure Inc.

Jan 2026 – July 2026

AI Engineer Intern

Bangalore, India

- Developed and ran a **seven-dimension LLM evaluation framework** and **six-criterion validation rubric** to reliability-test **agentic AI workflows**, **RAG pipelines**, and **prompt-engineering** pipelines for AccureIQx, an enterprise **generative AI** search and reasoning platform.
- Constructed **5 adversarial stress-test scenarios** using synthetic **cybersecurity** incident datasets with realistic log data to probe failure modes in **agentic reasoning** and **anomaly-detection** systems.
- Cut manual weekly reporting time by **35%**, saving analyst hours, by building automated **Python (Pandas, NumPy)** analytics pipelines integrating multi-source data via **REST APIs**.

Taskar – India's 1st Healthcare Mall

May 2025 – July 2025

Software Developer Intern

Remote, India

- Reduced manual reporting workflows by **40%**, cutting process time, by building **microservice-based RESTful API** integrations to retrieve and analyze live **healthcare and inventory data** for business stakeholders.
- Improved **analytics dashboard** load times by **30%**, per page-render benchmarks, by optimizing **UI rendering** and **asynchronous data fetching** with **React Hooks**.
- Increased dashboard user engagement by **15%**, confirmed via internal **A/B test** results, by redesigning dashboard layouts based on stakeholder feedback.

Research Designs and Standards Organisation (R.D.S.O)

May 2024 – July 2024

Software Engineering Intern

Lucknow, India

- Load-tested a responsive, **service-oriented attendance tracking system (HTML, CSS, JavaScript, PHP, MySQL)** to sustain **500+ concurrent users**, using **Apache JMeter** scalability and stress testing across peak-usage scenarios.
- Cut authentication response times by **60%** for **1,000+ daily users** by redesigning the login workflow and adding **AJAX-based** real-time interactivity.

## Projects

CFR Compliance Checker: Agentic Contract Auditor | *Python, FastMCP, Agno, FastAPI, PostgreSQL* | [Github](#)

- Architected an **agentic compliance pipeline** (security → **eCFR retrieval** → deterministic rules → **LLM reasoning** → **verification** → human review) with **evidence provenance** attached only by the retrieval layer, routing unresolved cases to **NEEDS REVIEW** instead of silent acceptance.
- Implemented a **human-in-the-loop review** lifecycle with **optimistic concurrency** and an immutable audit trail, plus an optional **PostgreSQL** backend with versioned migrations, verified across **268 offline tests**.

Incident Response Engine | *Python, Agno, FastMCP, Kafka, ChromaDB, Scikit-learn* | [Github](#)

- Engineered a **Kafka** event pipeline gating an LLM agent pipeline behind a deterministic z-score filter and **ML anomaly gate (IsolationForest + LOF)**, live-validated end-to-end against a real broker and sandboxed **MCP** diagnostics.
- Designed a **two-phase Investigator Agent** separating tool-calling from analysis to prevent self-reported tool-use, and enforced a hard **human-approval gate (Slack + dashboard HITL)** before any remediation, proven through a **220-test suite** and a live tool-calling run.

Smart ATS: Agentic GenAI Career Copilot | *Python, FastAPI, LangGraph, MCP, PostgreSQL, ChromaDB, Kafka* | [Github](#)

- Built a deterministic **Hybrid ATS** (semantic + keyword + skill-graph, explainable scoring) as the source of truth, with a 9-node **LangGraph** Career Copilot and an 11-tool **MCP** server that explain scores and generate grounded suggestions **without ever fabricating a skill**.
- Benchmarked on a 30-pair hand-labeled evaluation suite (Spearman **0.839** vs. human rank, Recommendation Recall@3 **0.913**), backed by **240 tests**, JWT auth, rate limiting, Prometheus metrics, and Kafka-based async ingestion.

## Technical Skills and Interests

**Languages:** Python, SQL, C++, JavaScript, HTML/CSS

**ML/DL Frameworks:** Scikit-learn, NumPy, Pandas, Matplotlib, OpenCV

**LLM/Agentic Stack:** LangChain, LangGraph, Agno, MCP, RAG

**Backend & Infrastructure:** FastAPI, Streamlit, PostgreSQL, ChromaDB (vector DB), Docker, Kafka

**Tools:** Git, Tableau

**AI/ML Domains:** Generative AI, LLMs, RAG (Retrieval-Augmented Generation), Agentic AI, NLP, Machine Learning, Anomaly Detection, Prompt Engineering, Model Evaluation, Data Engineering, Data Analysis